

IPL CRUNCH '26

Data Analytics Case Study

1,218

Matches

289,673

Balls Analyzed

22

Visualizations

18

Seasons

Built with Python | Pandas | Matplotlib | Seaborn | Scikit-learn

HTML/CSS/JS Interactive Dashboard

Executive Summary

The Problem

Every IPL season, 500M+ viewers hear recycled opinions: 'Toss is everything,' 'Batting wins T20s,' 'Middle overs don't matter.' But statistical evidence behind these claims is almost nonexistent. Analysts quote win percentages without running hypothesis tests.

Our Approach

We analyzed 289,673 ball-by-ball deliveries across 1,218 IPL matches spanning 18 seasons (2007/08-2026). We applied Chi-squared hypothesis testing, Logistic Regression modeling, and deep phase-wise analysis to separate fact from folklore.

Key Findings

1. TOSS MYTH DEBUNKED: 50.5% win rate for toss winners ($p=0.731$) - statistically no better than a coin flip. 2. DEATH OVERS DOMINATE: 47.3% importance vs Middle Overs at just 6.5% - a 7.3x gap. 3. HIDDEN PREDICTOR: Winning teams have a 42% higher wicket-taking rate in death overs. Death-over bowling, not batting, is the strongest match-winning skill in the IPL.

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn, HTML/CSS/JS Dashboard

Methodology

Data Ingestion & Cleaning

Loaded ball-by-ball CSV from Cricsheet (289,673 records). Normalized 20+ historical franchise names into 15 current entities (e.g., Deccan Chargers -> Sunrisers Hyderabad). Engineered features: match phases (Powerplay: overs 1-6, Middle: 7-15, Death: 16-20), ball event flags, and match-level aggregations.

Statistical Testing

Applied Chi-squared test of independence to evaluate toss-win correlation. H_0 : Toss outcome and match outcome are independent. Result: $\chi^2=0.12$, $p=0.731$ - failed to reject null hypothesis. The toss provides no statistically significant advantage.

Predictive Modeling

Trained a Logistic Regression classifier using phase-wise run totals (Powerplay, Middle, Death) as features and match outcome (win/loss) as target. Model coefficients converted to relative importance weights revealed Death (47.3%) and Powerplay (46.2%) as near-equal predictors, with Middle overs (6.5%) being negligible.

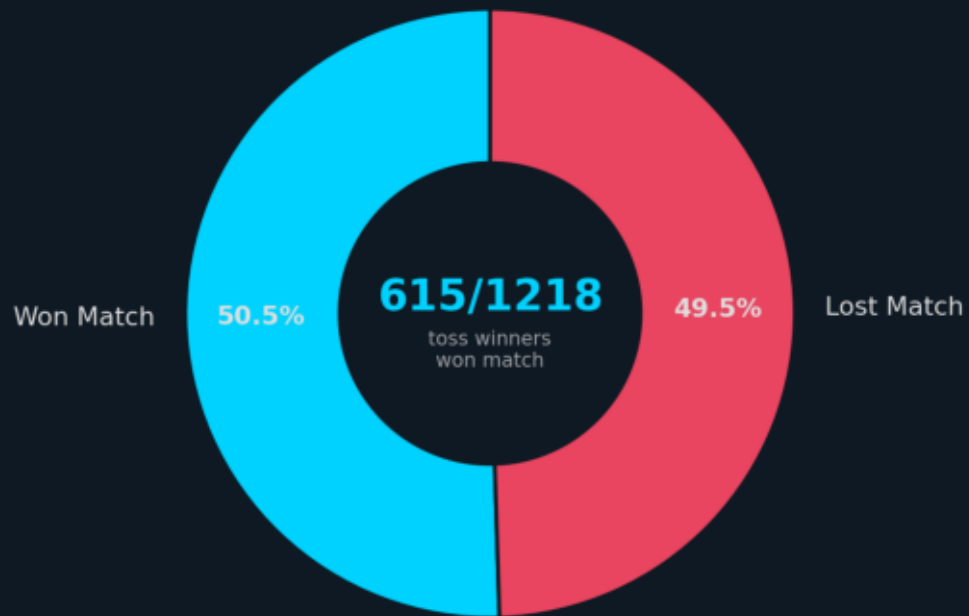
Visualization & Dashboard

Generated 22 publication-quality charts using Matplotlib and Seaborn with a consistent dark theme. Built a 5-tab interactive web dashboard using vanilla HTML, CSS, and JavaScript with glassmorphic design, animated counters, and scroll-driven reveal animations.

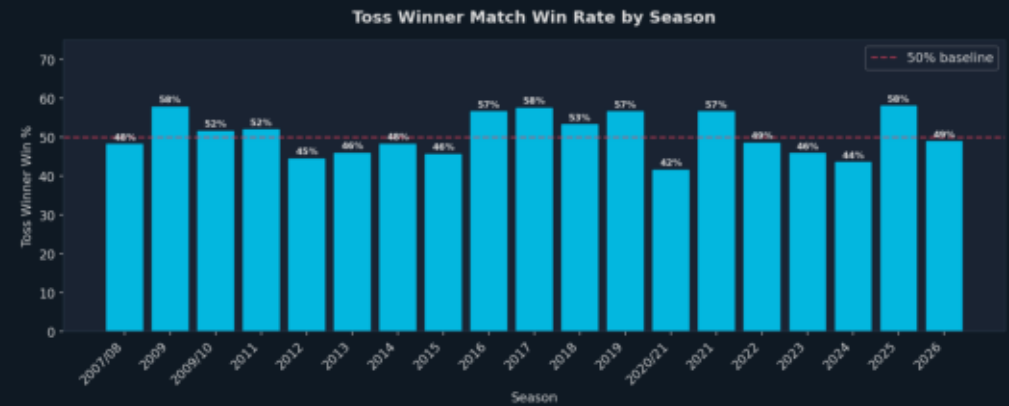
Finding 1: Toss Impact Analysis

Toss win -> match win: 50.5% | Chi-squared $p=0.731$ (NOT significant)

Does Winning the Toss = Winning the Match?



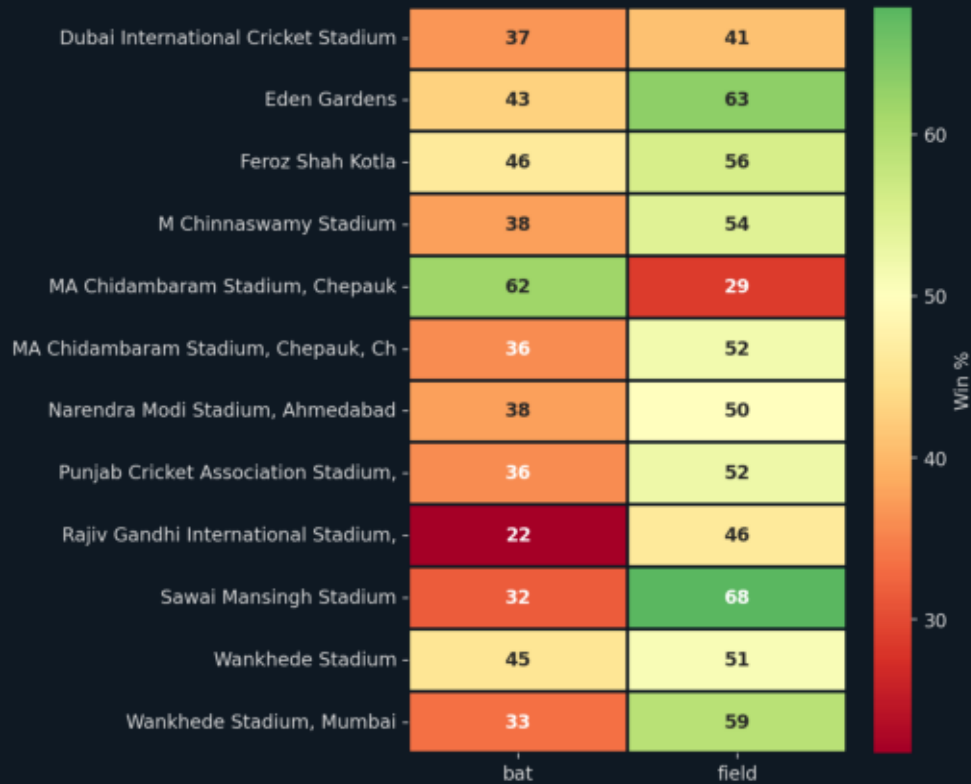
$\chi^2 = 0.12, p = 0.7310$



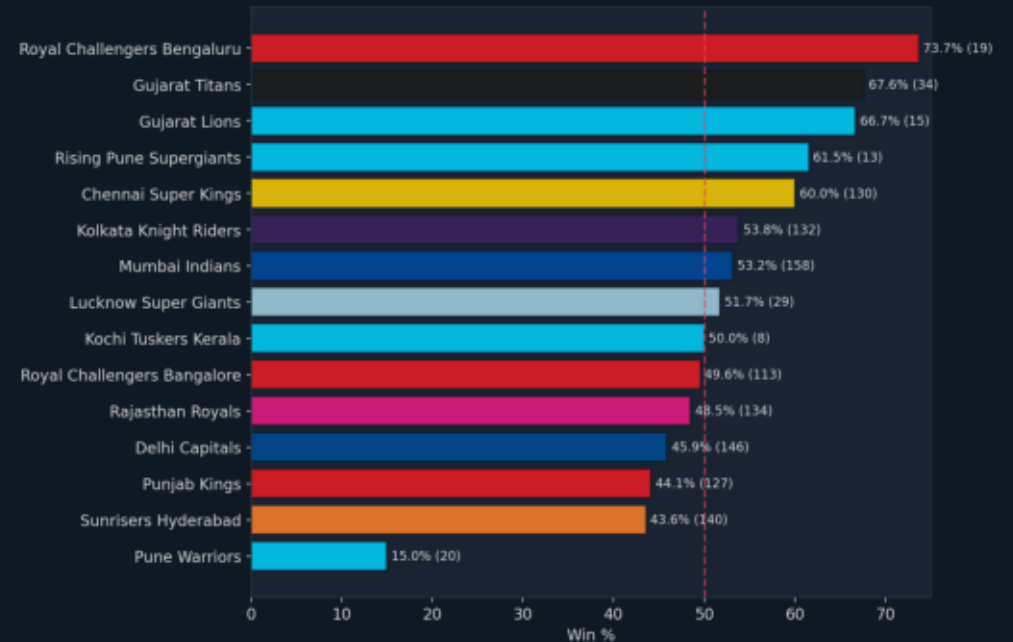
Toss Impact: Venue & Team Breakdown

Some venues show slight bias, but no team consistently capitalizes on the toss

Toss Winner Win % by Venue & Decision

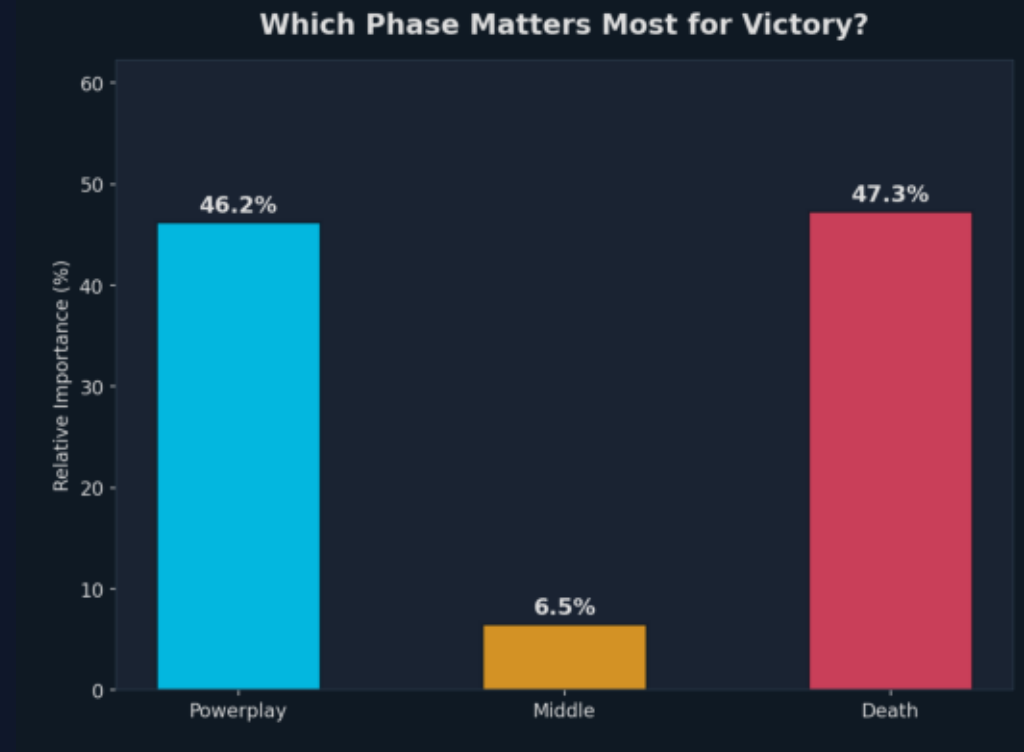
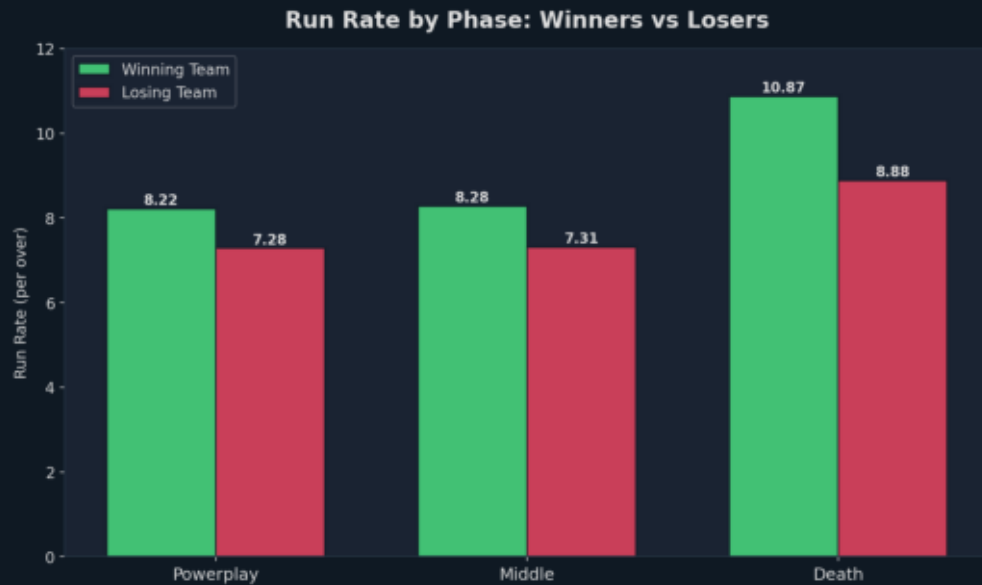


Toss Win → Match Win % by Team



Finding 2: Phase-wise Match Impact

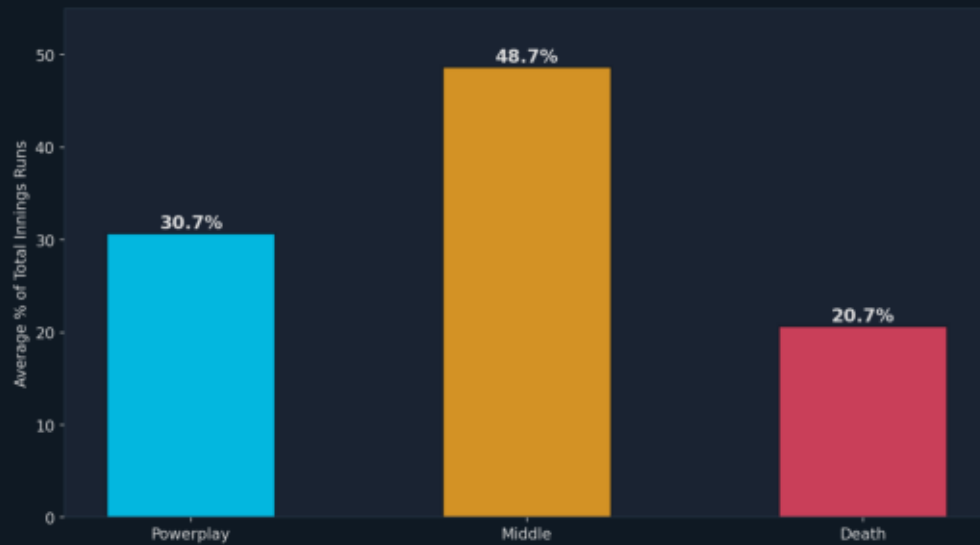
Death overs (47.3%) and Powerplay (46.2%) are nearly equally decisive



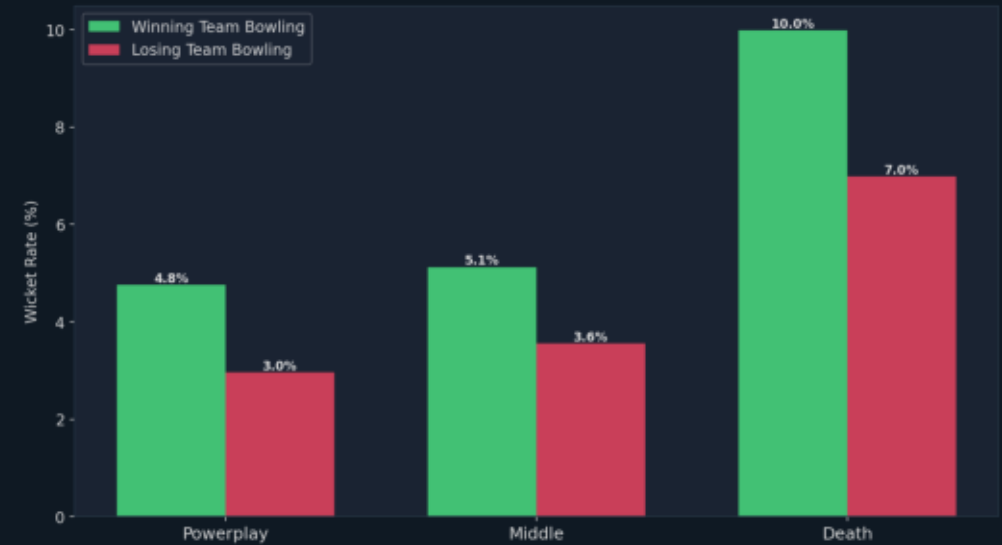
Phase Analysis: Run Contribution & Wickets

Winners dominate run rate in Powerplay and take more wickets at the death

Run Contribution by Phase



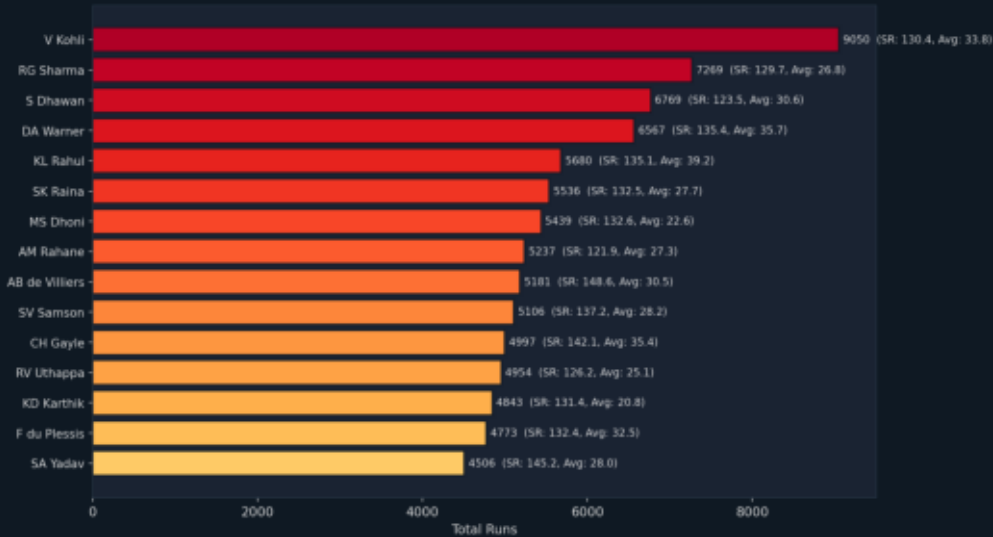
Bowling Wicket Rate by Phase: Winners vs Losers



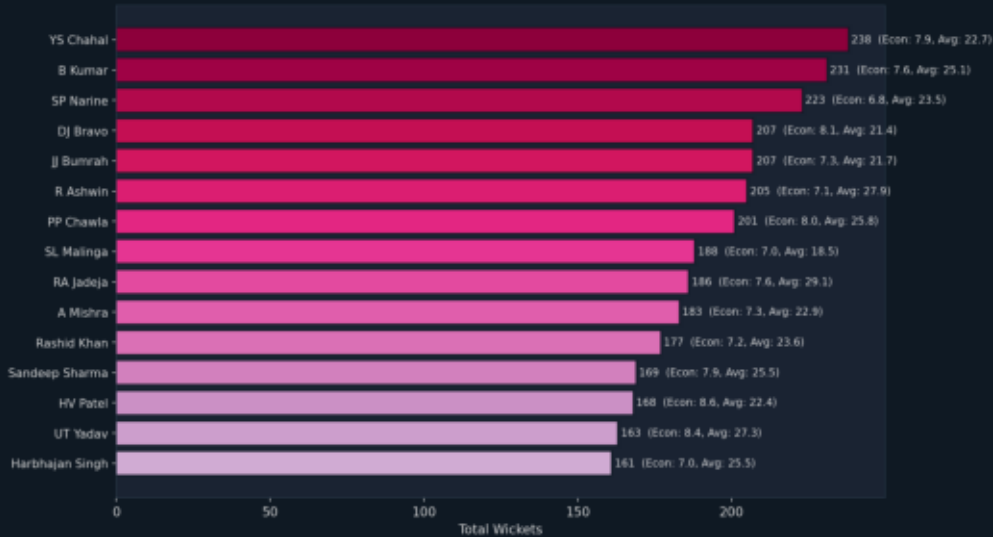
Finding 3: Top Performers

V Kohli leads with 9,050 runs | YS Chahal leads with 238 wickets

Top 15 Run Scorers (min 20 innings)



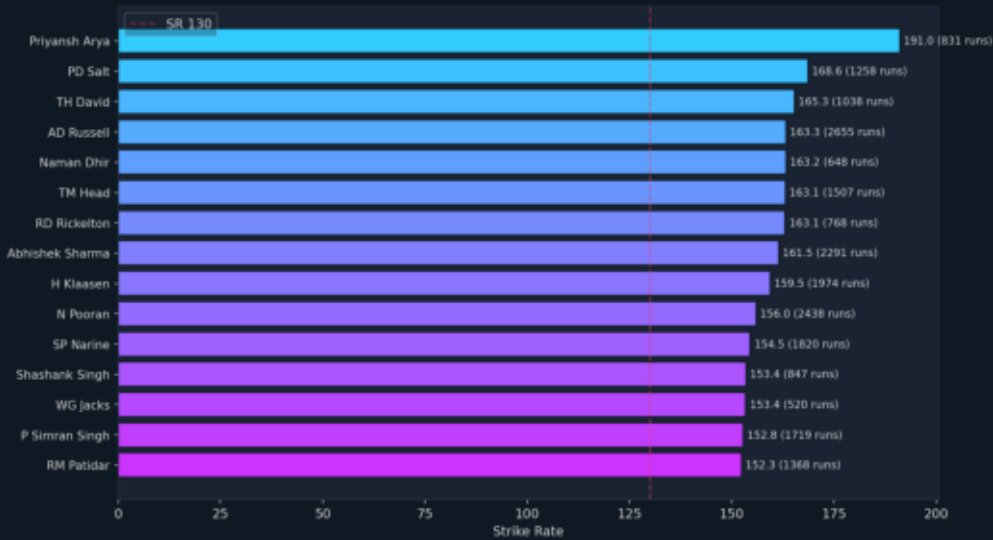
Top 15 Wicket Takers (min 20 innings)



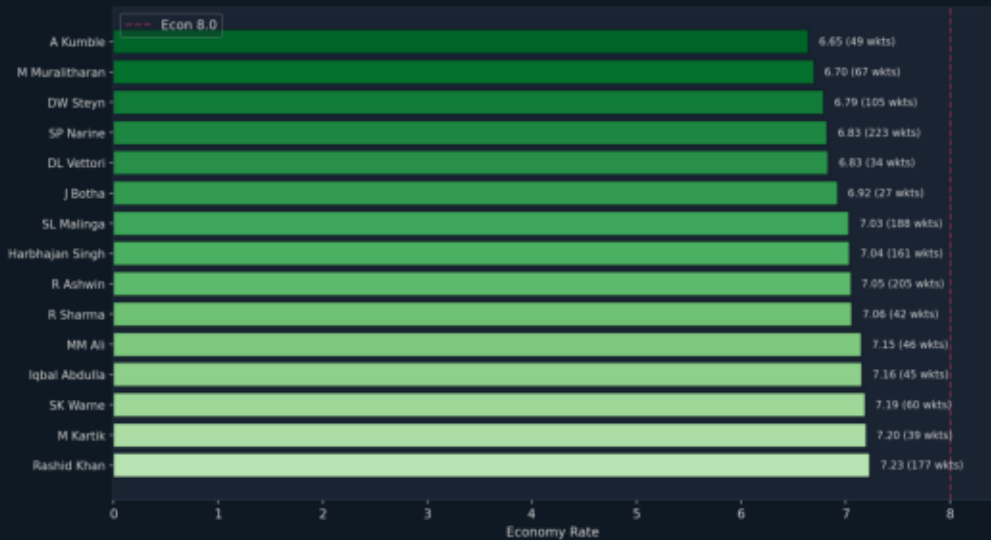
Performance Deep Dive

Strike rates and economy rates reveal the specialists

Highest Strike Rates (min 500 runs)



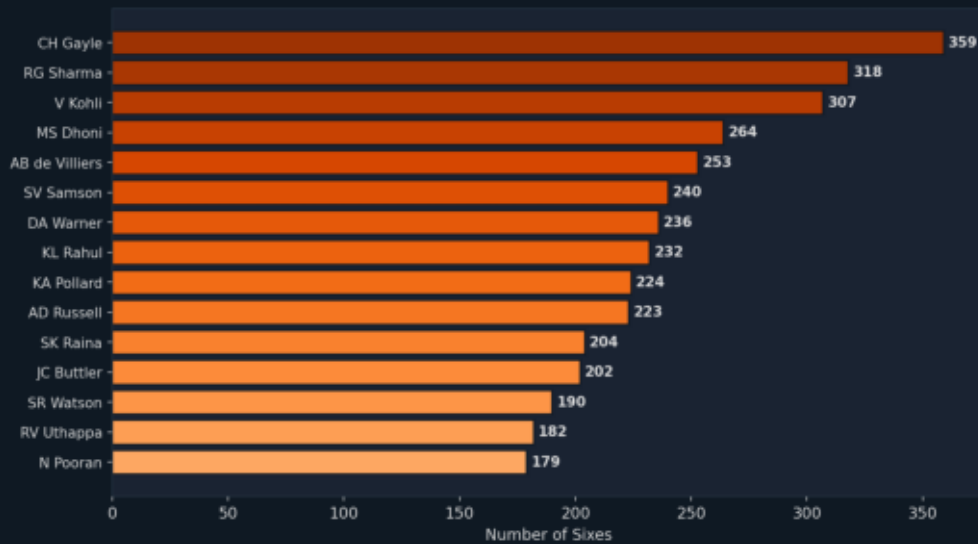
Best Economy Rates (min 30 innings)



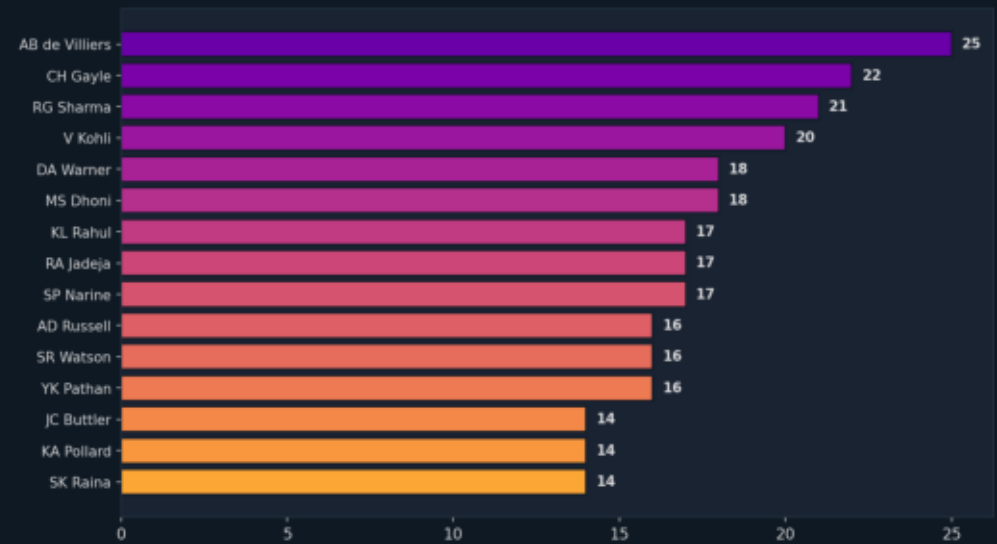
Power Hitters & Match Winners

CH Gayle: 359 sixes | AB de Villiers: 25 Player of the Match awards

Most Sixes in IPL History

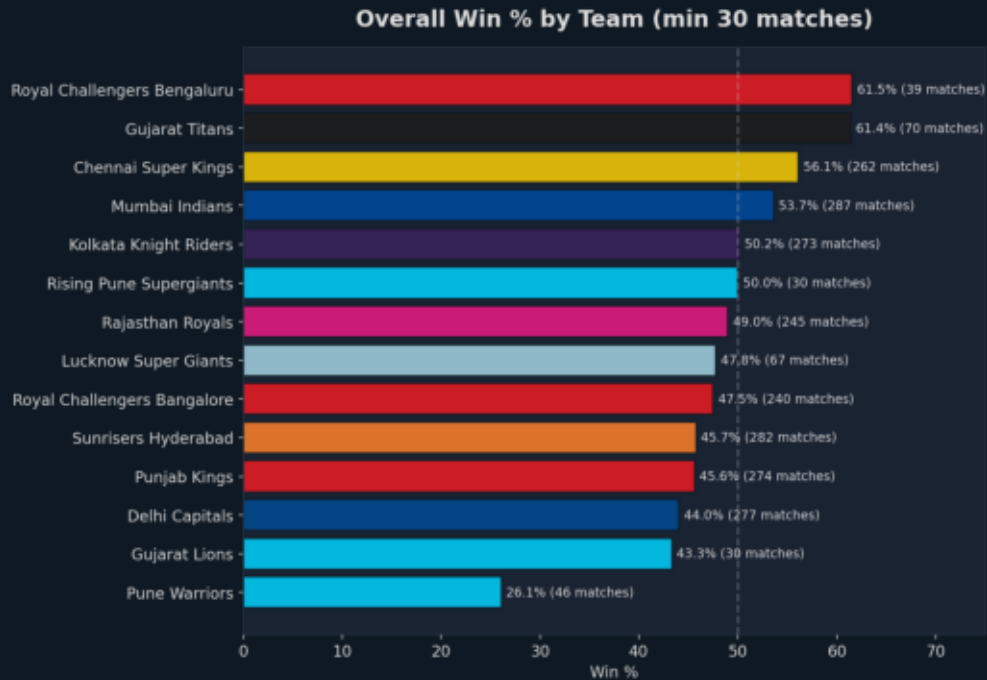
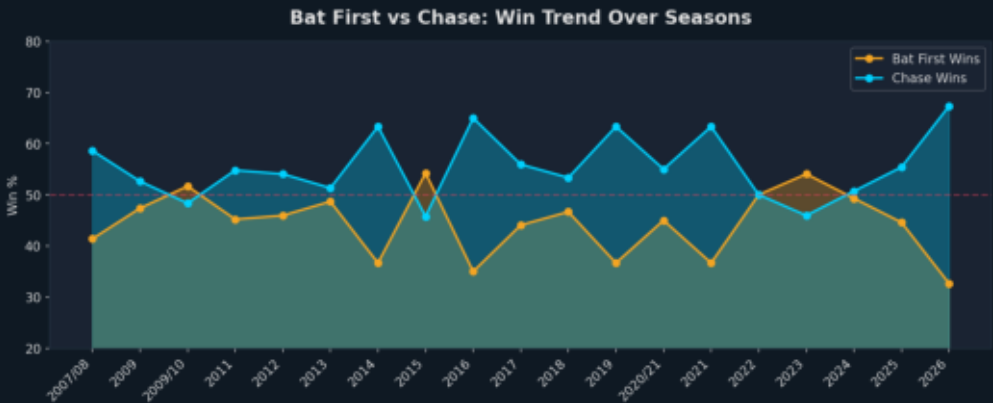


Most Player of the Match Awards



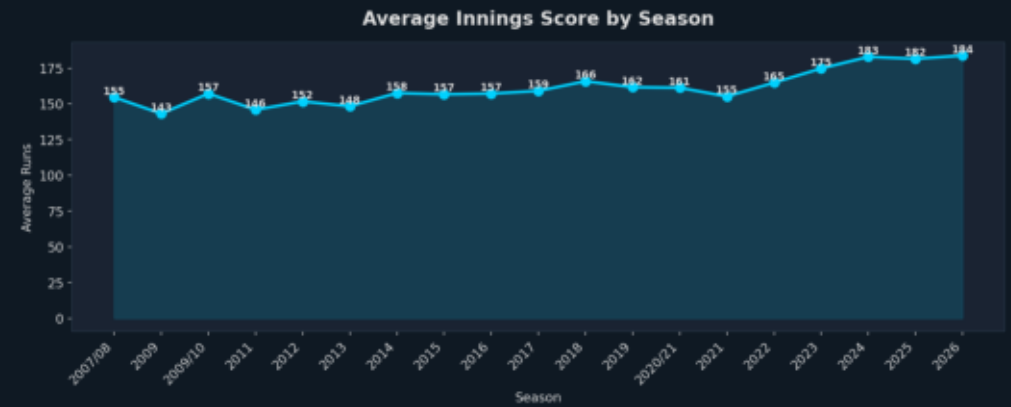
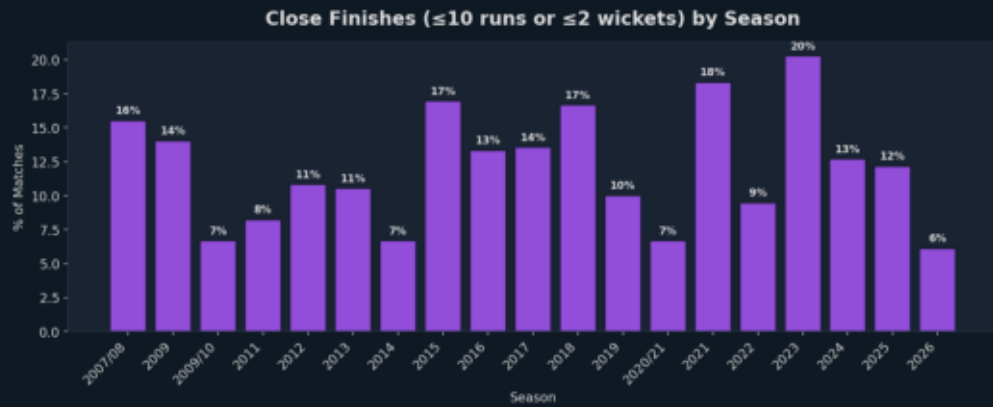
Finding 4: Hidden Patterns

Chasing has become increasingly dominant across seasons



Deeper Insights

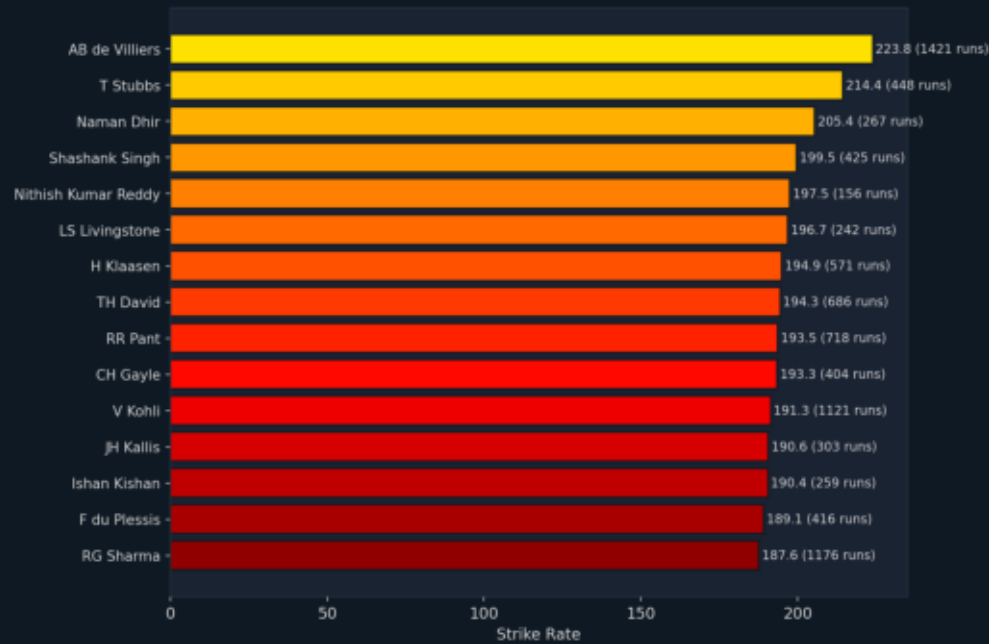
Average innings score has risen from 155 (2007/08) to 184 (2026)



Finding 5: The Surprise - Death Over Bowling

Death-over BOWLING is the most underrated match-winning skill in the IPL

Death Over Specialists — Highest SR (overs 16-20, min 15 inn)

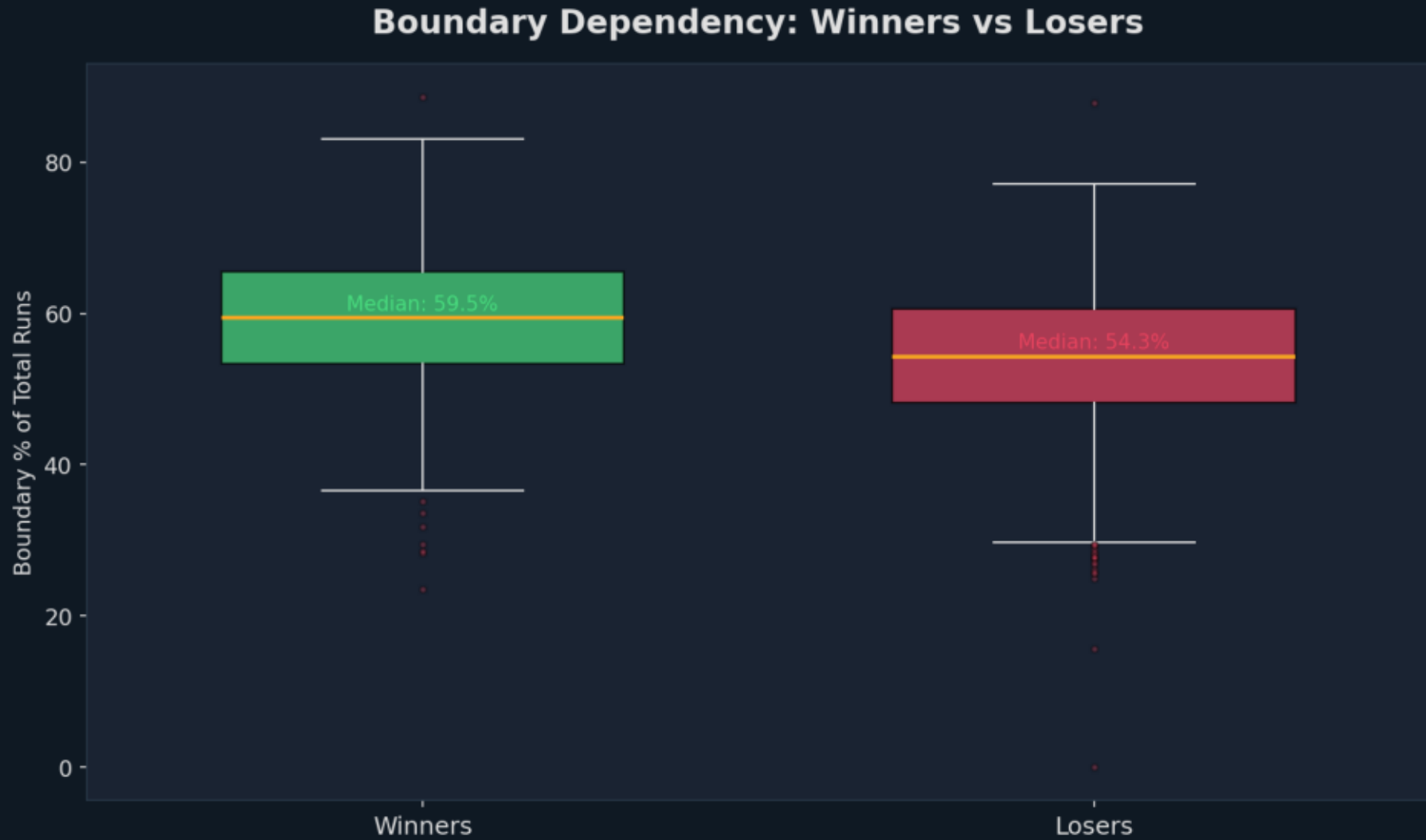


Powerplay Kings — Best Economy (overs 1-6, min 15 inn)



Boundary Dependency Analysis

Winners score 59.5% of runs from boundaries vs 54.3% for losers



Conclusion & Key Takeaways

1. Toss is a Myth

At 50.5% ($p=0.731$), the toss advantage is statistically indistinguishable from a coin flip. Captains and commentators vastly overestimate its importance.

2. Death & Powerplay Decide Everything

Together, Death overs (47.3%) and Powerplay (46.2%) account for 93.5% of the predictive weight in determining match outcomes. Middle overs are nearly irrelevant at 6.5%.

3. Death-Over Bowling is the Hidden Gem

The biggest surprise: winning teams maintain a 42% higher wicket-taking rate in the death overs. In an era obsessed with batting firepower, the data proves that elite death bowling is what truly separates winners from losers.

4. Run Inflation is Real

Average innings scores have risen from 155 in 2007/08 to 184 in 2026, a 19% increase across 18 seasons. Teams defending totals must adapt with aggressive bowling strategies.

This analysis was performed on 289,673 deliveries across 1,218 matches spanning 18 IPL seasons, using Python, statistical hypothesis testing, and machine learning.